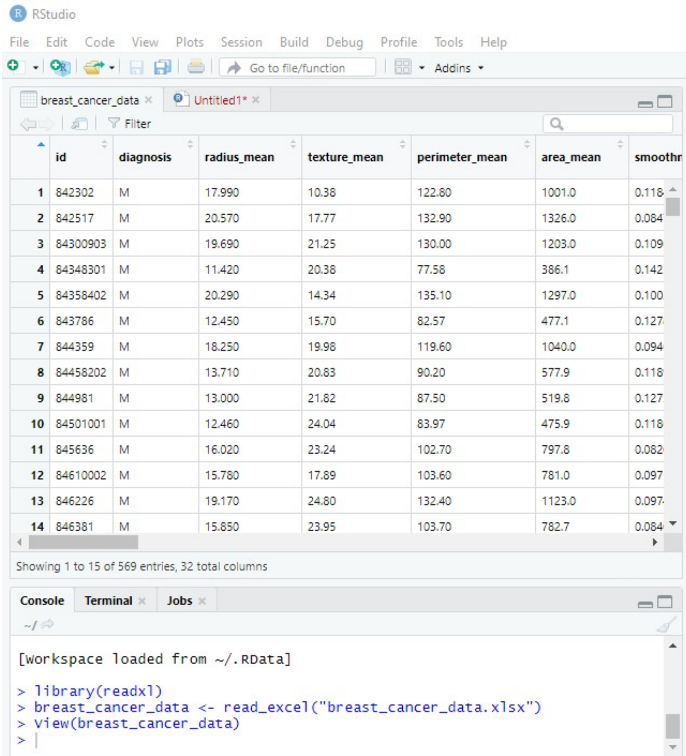
Week 8 Sean Graham

1.

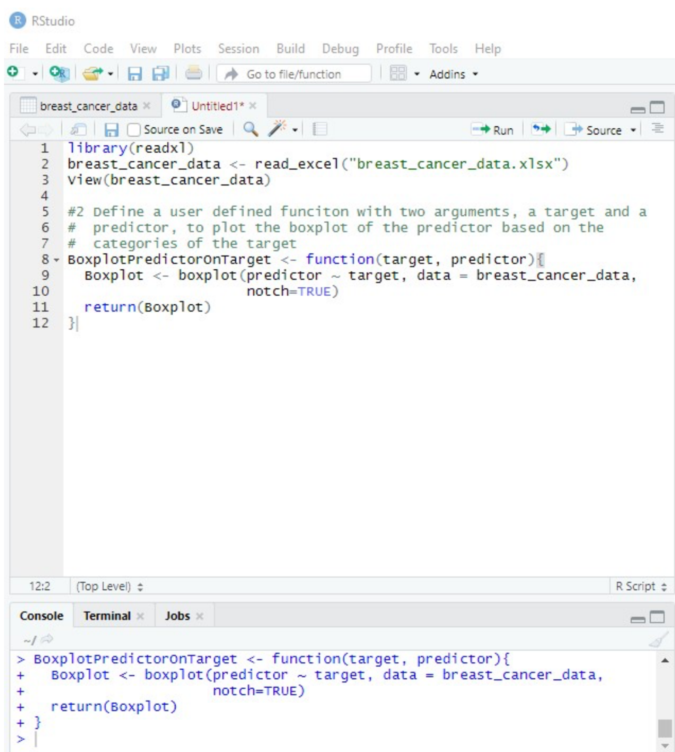


	id	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean	smoothn
1	842302	M	17.990	10.38	122.80	1001.0	0.118
2	842517	M	20.570	17.77	132.90	1326.0	0.084
3	84300903	M	19.690	21.25	130.00	1203.0	0.109
4	84348301	M	11.420	20.38	77.58	386.1	0.142
5	84358402	M	20.290	14.34	135.10	1297.0	0.100
6	843786	M	12.450	15.70	82.57	477.1	0.127
7	844359	M	18.250	19.98	119.60	1040.0	0.094
8	84458202	M	13.710	20.83	90.20	577.9	0.118
9	844981	M	13.000	21.82	87.50	519.8	0.127
10	84501001	M	12.460	24.04	83.97	475.9	0.118
11	845636	M	16.020	23.24	102.70	797.8	0.082
12	84610002	M	15.780	17.89	103.60	781.0	0.097
13	846226	M	19.170	24.80	132.40	1123.0	0.097
14	846381	M	15.850	23.95	103.70	782.7	0.084


```
[Workspace loaded from ~/.RData]

> library(readxl)
> breast_cancer_data <- read_excel("breast_cancer_data.xlsx")
> View(breast_cancer_data)
>
```

2.



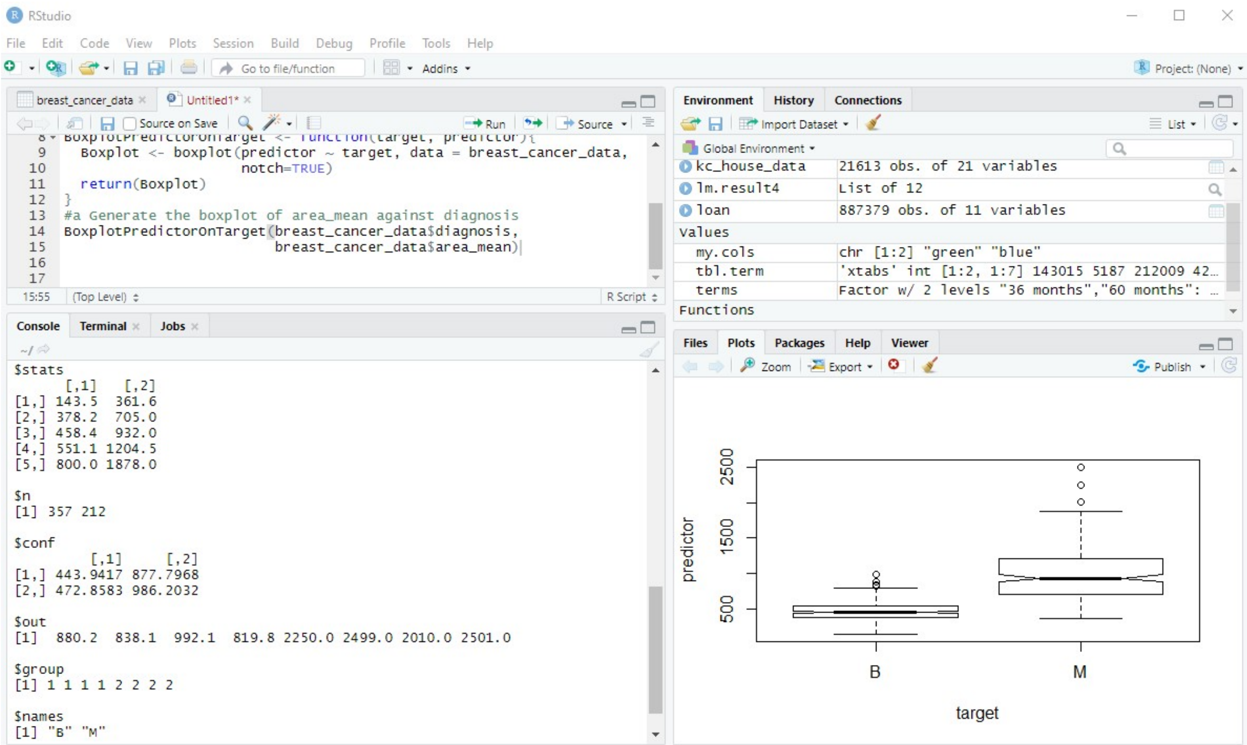

```
library(readxl)
breast_cancer_data <- read_excel("breast_cancer_data.xlsx")
View(breast_cancer_data)

#2 Define a user defined funciton with two arguments, a target and a
#  predictor, to plot the boxplot of the predictor based on the
#  categories of the target
BoxplotPredictorOnTarget <- function(target, predictor){
  Boxplot <- boxplot(predictor ~ target, data = breast_cancer_data,
                     notch=TRUE)
  return(Boxplot)
}
```



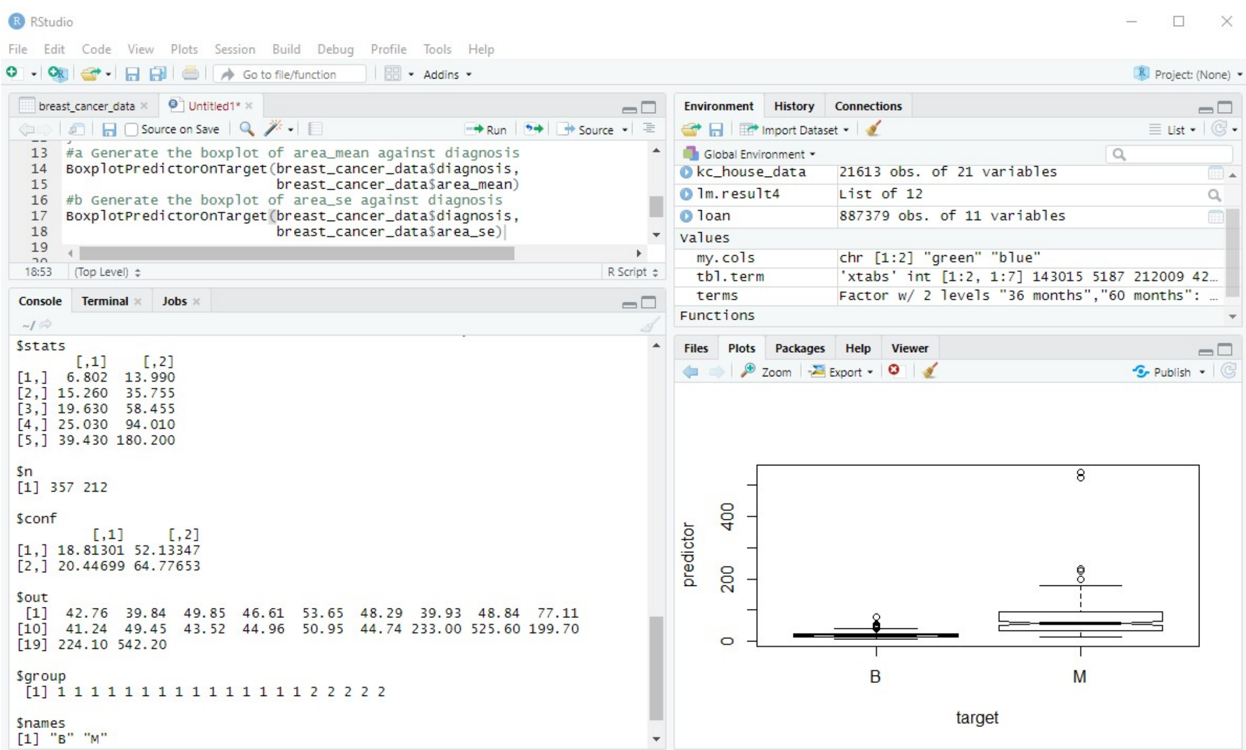
```
> BoxplotPredictorOnTarget <- function(target, predictor){
+   Boxplot <- boxplot(predictor ~ target, data = breast_cancer_data,
+                      notch=TRUE)
+   return(Boxplot)
+ }
>
```

2a.



RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
breast_cancer_data
Untitled1*
Source on Save
Run
Source
BoxplotPredictorOnTarget <- function(target, predictor){
Boxplot <- boxplot(predictor ~ target, data = breast_cancer_data,
notch=TRUE)
return(Boxplot)
}
#a Generate the boxplot of area_mean against diagnosis
BoxplotPredictorOnTarget(breast_cancer_data$diagnosis,
breast_cancer_data$area_mean)
15:55
(Top Level)
R Script
Console
Terminal
Jobs
$stats
[,1] [,2]
[1,] 143.5 361.6
[2,] 378.2 705.0
[3,] 458.4 932.0
[4,] 551.1 1204.5
[5,] 800.0 1878.0
$n
[1] 357 212
$conf
[,1] [,2]
[1,] 443.9417 877.7968
[2,] 472.8583 986.2032
$out
[1] 880.2 838.1 992.1 819.8 2250.0 2499.0 2010.0 2501.0
$group
[1] 1 1 1 1 2 2 2 2
$names
[1] "B" "M"
Environment
History
Connections
Import Dataset
Global Environment
kc_house_data
21613 obs. of 21 variables
lm.result4
List of 12
loan
887379 obs. of 11 variables
Values
my.cols
chr [1:2] "green" "blue"
tbl.term
'xtabs' int [1:2, 1:7] 143015 5187 212009 42...
terms
Factor w/ 2 levels "36 months","60 months": ...
Functions
Files
Plots
Packages
Help
Viewer
Zoom
Export
Publish
predictor
500
1500
2500
B
M
target

2b.



RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
breast_cancer_data
Untitled1*
Source on Save
Run
Source
#a Generate the boxplot of area_mean against diagnosis
BoxplotPredictorOnTarget(breast_cancer_data$diagnosis,
breast_cancer_data$area_mean)
#b Generate the boxplot of area_se against diagnosis
BoxplotPredictorOnTarget(breast_cancer_data$diagnosis,
breast_cancer_data$area_se)
18:53
(Top Level)
R Script
Console
Terminal
Jobs
$stats
[,1] [,2]
[1,] 6.802 13.990
[2,] 15.260 35.755
[3,] 19.630 58.455
[4,] 25.030 94.010
[5,] 39.430 180.200
$n
[1] 357 212
$conf
[,1] [,2]
[1,] 18.81301 52.13347
[2,] 20.44699 64.77653
$out
[1] 42.76 39.84 49.85 46.61 53.65 48.29 39.93 48.84 77.11
[10] 41.24 49.45 43.52 44.96 50.95 44.74 233.00 525.60 199.70
[19] 224.10 542.20
$group
[1] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 2 2 2
$names
[1] "B" "M"
Environment
History
Connections
Import Dataset
Global Environment
kc_house_data
21613 obs. of 21 variables
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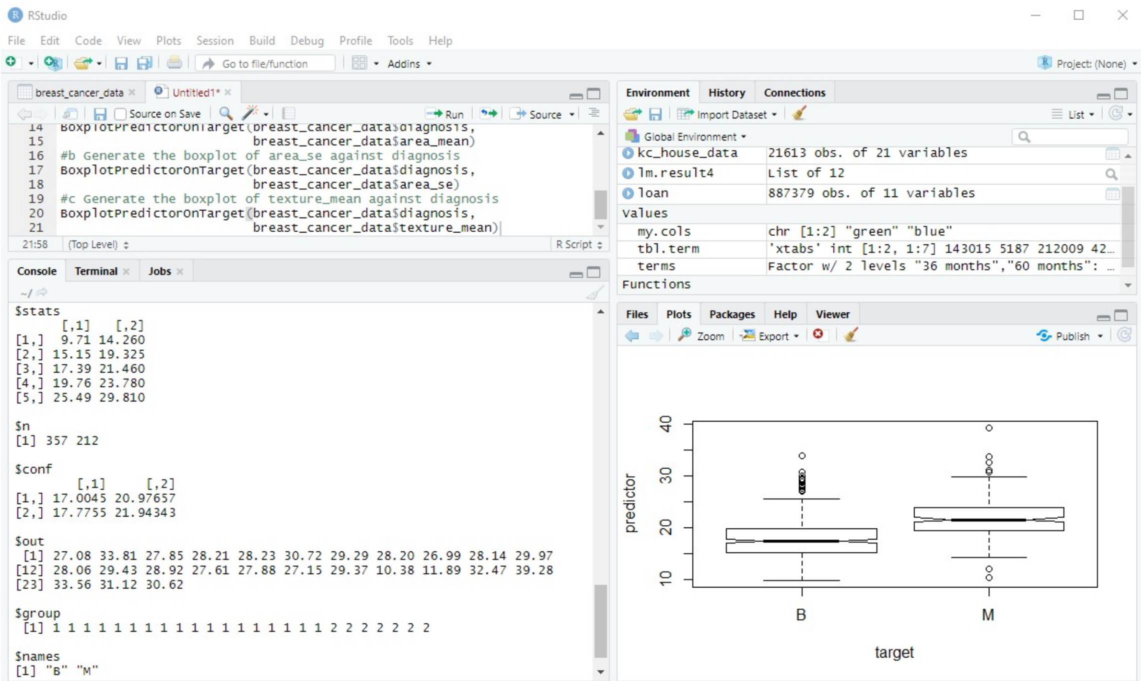
terms
Factor w/ 2 levels "36 months","60 months": ...
Functions
Files
Plots
Packages
Help
Viewer
Zoom
Export
Publish
predictor
0
200
400
B
M
target

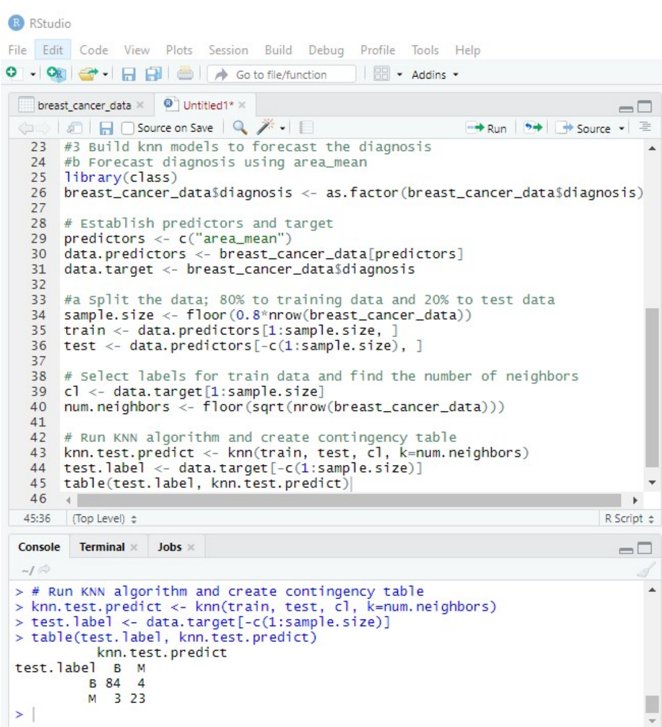
2c.



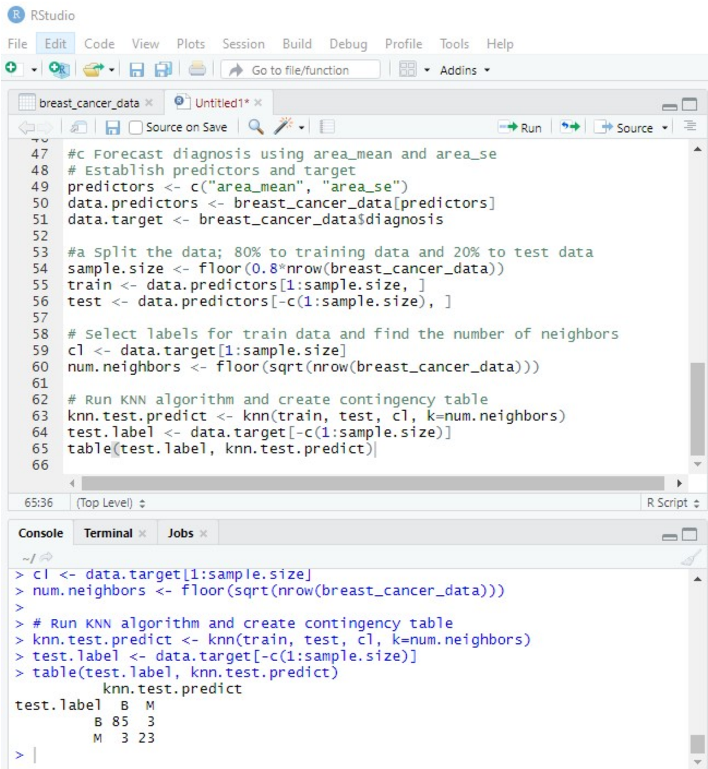
3. Accuracy for B: (84+23)/(84+32+4+3) = 93.86% Accuracy for C and D: (85+23)/(85+23+3+3) = 94.74%
C and D are the best models by accuracy (both have the same numbers, and thus the same accuracy).

3a. This is performed within parts b, c, and d. Check the codes in the subsequent images.

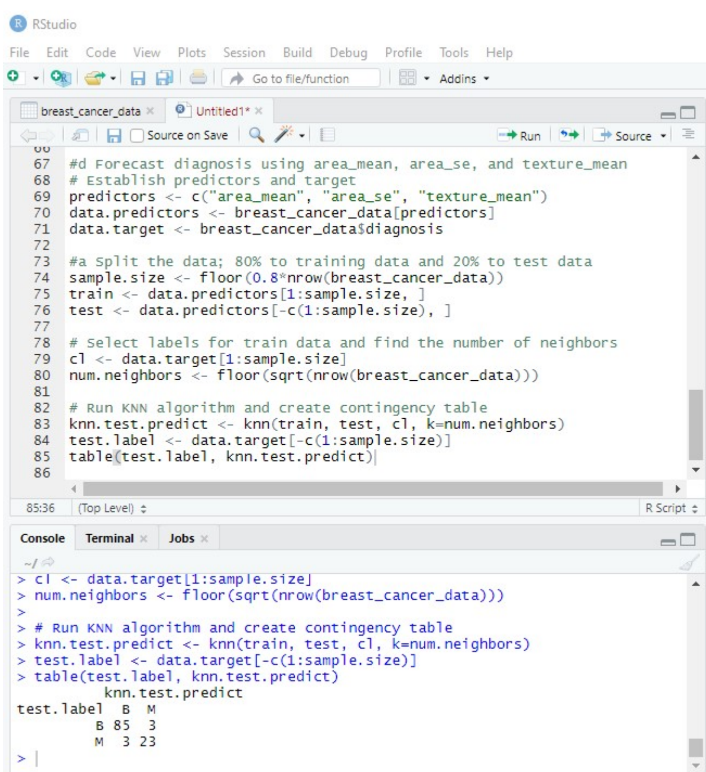
3b.



3c.


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
breast_cancer_data
Untitled1*
Source on Save
Run
Source
#c Forecast diagnosis using area_mean and area_se
# Establish predictors and target
predictors <- c("area_mean", "area_se")
data.predictors <- breast_cancer_data[predictors]
data.target <- breast_cancer_data$diagnosis
#a Split the data; 80% to training data and 20% to test data
sample.size <- floor(0.8*nrow(breast_cancer_data))
train <- data.predictors[1:sample.size,]
test <- data.predictors[-c(1:sample.size),]
# Select labels for train data and find the number of neighbors
cl <- data.target[1:sample.size]
num.neighbors <- floor(sqrt(nrow(breast_cancer_data)))
# Run KNN algorithm and create contingency table
knn.test.predict <- knn(train, test, cl, k=num.neighbors)
test.label <- data.target[-c(1:sample.size)]
table(test.label, knn.test.predict)
65:36
(Top Level)
R Script
Console
Terminal
Jobs
~/
> cl <- data.target[1:sample.size]
> num.neighbors <- floor(sqrt(nrow(breast_cancer_data)))
>
> # Run KNN algorithm and create contingency table
> knn.test.predict <- knn(train, test, cl, k=num.neighbors)
> test.label <- data.target[-c(1:sample.size)]
> table(test.label, knn.test.predict)
knn.test.predict
test.label B M
B 85 3
M 3 23
>

3d.


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Addins
breast_cancer_data
Untitled1*
Source on Save
Run
Source
#d Forecast diagnosis using area_mean, area_se, and texture_mean
# Establish predictors and target
predictors <- c("area_mean", "area_se", "texture_mean")
data.predictors <- breast_cancer_data[predictors]
data.target <- breast_cancer_data$diagnosis
#a Split the data; 80% to training data and 20% to test data
sample.size <- floor(0.8*nrow(breast_cancer_data))
train <- data.predictors[1:sample.size,]
test <- data.predictors[-c(1:sample.size),]
# Select labels for train data and find the number of neighbors
cl <- data.target[1:sample.size]
num.neighbors <- floor(sqrt(nrow(breast_cancer_data)))
# Run KNN algorithm and create contingency table
knn.test.predict <- knn(train, test, cl, k=num.neighbors)
test.label <- data.target[-c(1:sample.size)]
table(test.label, knn.test.predict)
85:36
(Top Level)
R Script
Console
Terminal
Jobs
~/
> cl <- data.target[1:sample.size]
> num.neighbors <- floor(sqrt(nrow(breast_cancer_data)))
>
> # Run KNN algorithm and create contingency table
> knn.test.predict <- knn(train, test, cl, k=num.neighbors)
> test.label <- data.target[-c(1:sample.size)]
> table(test.label, knn.test.predict)
knn.test.predict
test.label B M
B 85 3
M 3 23
>